

WundCheck — Mini-CDSS for daily wound checks after surgery

FHNW MSc Medical Informatics • Clinical Decision Support Systems Module 2026

Group project: knowledge-based CDSS following the WHO SMART Guidelines layer approach (L1 → L2 → L3 → L4)

Repository: <https://github.com/Sivanajani/smart-wound-guideline-app>

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What it is about

Postoperative wound infections (Surgical Site Infections, SSI) predominantly occur **after discharge from hospital** — that is, precisely when no health professional looks at the wound on a daily basis any more. Patients often recognise warning signs too late, or call unnecessarily because of normal signs of healing.

WundCheck is a daily self-check: in 2–3 minutes, patients answer structured questions and select their wound appearance via **clickable graphics** instead of free text or a photo. A rule-based system classifies the result according to established SSI criteria into a traffic light and states specifically what needs to be done.

● GREEN	healing normal	check again tomorrow
● YELLOW	abnormal	observe, raise it at the next appointment
● ORANGE	suspected infection	contact the practice today
● RED	emergency	practice / emergency service immediately

The layer architecture

L1 Guidelines & Evidence Score	CDC/NHSN SSI criteria • NICE NG125 • Southampton Wound
	↓ human interpretation, documented decisions
L2 Human-readable Algorithm	BPMN workflow • Data Dictionary • Decision Table
	↓ translation into machine readability
L3 Machine-readable Layer	wundcheck-l3.json (elements, rules, constraints)
	↓ generic execution
L4 Executable Layer dynamically	single-file web app (vanilla JS), renders the L3

The central design decision: the app contains **no** clinical logic. Questions, visibility rules, calculations, thresholds and decision rules reside entirely in the L3 JSON. A change to a guideline is a change to *one file* — no rebuild, no developer.

Repository structure

Path	Content
docs/	The complete project documentation — one document per assignment item
l3/	Machine-readable CDSS definition (JSON) + schema documentation
l4/	Executable app — one self-contained HTML file, no build step, no dependencies
diagrams/	BPMN diagrams (<code>.bpmn</code> editable, <code>.svg</code> / <code>.png</code> for embedding)
assets/	Wound and discharge graphics for the image selection
submission/	Submission package: slides, demo recording, every document as PDF, L3 + L4
_archive/	Earlier file versions from before the restructuring

Documentation

The documents follow the structure of the assignment exactly:

#	Document	Assignment item
01	Health Need & Scope	Describe the health / health system need
02	Personas & Scenarios	User Personas and Scenarios
03	L1 — Guidelines & Evidence	Guidelines and Evidence
04	L2 — Workflow (BPMN)	L2: High-level workflow
05	L2 — Data Dictionary	L2: Data dictionary
06	L2 — Decision Logic	L2: Decision logic
07	L2 — L1→L2 challenges	Challenges in translation of L1 to L2
08	L3 — Architecture & Schema	Develop the machine-readable code
09	L4 — Implementation & UX	Implement the executable layer
10	QA — Risk-based testing	Quality Assurance Testing
11	Implementation, Monitoring & Evaluation	Implementation, monitoring and evaluation strategy
12	Expected Impact & Regulation	Describe expected impact
13	Project plan & contributions	Individual contributions
14	Presentation script	Spoken script for all 26 slides, timings, speaker split, live-demo choreography, Q&A map

Starting the app

```
# No server, no build, no installation
open l4/wundcheck-app.html      # macOS
start l4\wundcheck-app.html     # Windows
```

The app is bilingual (EN/DE, English by default) and loads the embedded L3. Via **"Load L3 file"** ("L3-Datei laden"), any other JSON can be loaded — which makes the separation of layers demonstrable live.

Running the tests

```
node tools/run-tests.mjs      # 19 decision-logic test cases against l3/wundcheck-l3.json
node tools/check-fhir.mjs     # 20 FHIR R4 conformance checks on the generated bundle
node tools/check-hardcoding.mjs # proves the L4 contains no clinical content
node tools/build-presentation.js # rebuilds the 26-slide deck incl. speaker notes and QR
code
python3 tools/qr_test.py      # validates the QR encoder (RS vector, BCH, round-trip)
python3 tools/embed-l3.py     # re-embed the L3 into the app after an L3 change
```

The app has **no build step and no dependencies**. Editing the L3 changes the CDSS; editing the HTML changes the rendering. Nothing else is required.

Traceability

Every artefact carries the same IDs. A rule can be traced back seamlessly:

DE-xxx	Data element	→ Data Dictionary	→ L3 elements[]	→ app question
R-xx	Decision Rule	→ Decision Table	→ L3 decision_rules[]	→ traffic-light output
C-xx	L1→L2 challenge	→ Challenges doc	→ affected rule	
H-xx	Hazard	→ Risk analysis	→ test case	
T-xx	Test case	→ Test protocol	→ bug ID	
B-xx	Bug	→ Bug log	→ fix → retest	

Scope

Covered: surgical wounds with primary wound closure (sutures, staples, tissue adhesive) in adults after elective procedures, **any body region**.

Not covered: chronic wounds, burns, secondarily healing and open wounds, children, contaminated emergency procedures.

 **Study prototype.** Not for clinical use. WundCheck does not replace medical assessment — it triages and documents.